

IAM

Identity and Access management architecture is the framework that helps securely manage user identities, authentication, & authorization across an organization's IT environment.

IAM Life Cycle

Identity (Start)



Provisioning



Authentication / Authorization



Self-services / Delegation



Password Management



Role & Application Access Management



Compliance Identity Certification & Reporting



De-Provisioning



Identity (End)

Key components of IAM Architecture

- Identity Management (Creating a digital ID).
- Authentication include:-
 - Single Sign-on (SSO)
 - Multi-Factor Authentication (MFA)
 - Password-less Authentication.
 - Adaptive Authentication.
- Authorization & Permission Controls.
- Access policies.
- Monitoring & Auditing.
- Integration with cloud services.

Core Components of IAM

IAM User

- * Represents a specific person or application with long-term credentials.
- * Permanent credentials.
- * Rarely used for direct access.

IAM Group

- * A logical container to simplify permission management across users.
- * This is not an identity.
- * Assigns common permissions
Ex:- Developers, Auditors.

JAM role

- * A temporary identity assumed by trusted entities for specific tasks.
 - * Temporary credentials
 - * Best practice for service-to-service access.
- Ex:- EC2 → S3.

Types of permissions

- Allow Permissions
- Deny Permissions. (overrides any allow rule to block access)
- Implicit Deny. (Default state).

Access Control (RBAC)

- * RBAC = Role Based Access Control
authorizes users based on predefined roles.
- * Permissions are assigned to roles,
& users are assigned to those roles.
- * RBAC works best for organizations with stable role definitions and predictable access needs.

Levels of RBAC

- Traditional RBAC ⇒ Users can accumulate roles over time, gradually expanding their permissions.

- Constrained RBAC ⇒ Puts limits on how many roles a user can have at once to prevent over permissioning
- Hierarchical RBAC ⇒ Aligns access with organizational rank, executive get broader access.

ABAC (Attribute-based access control)

* This governs authorization through dynamic rules that go beyond simple roles

* Decisions are made ~~of~~ based on user, resource & environmental attributes to determine access.

Three categories of Attributes

- User & Action attributes.
- Asset attributes.
- Environmental context

* This best suited for large or complex organizations, highly regulated industries

Policy Based Access control (PBAC)

- * This uses policies to manage identities respective access, dynamic ownership.
- * Automates IAM, IGA processes.

* IAM is the cornerstone of Trust, Reputation & Risk management in the digital era.

Authorization

* This is the process of granting or denying access to specific resources or functionalities based on the authenticated user's permissions.

Authentication

* This is the process of validating that users are whom they claim to be.

* In simple words it ensures that the user's identity is genuine or not.

→ Authentication precedes authorization.

Authentication Methods

→ Password - Based Authentication.

→ Physical Identification

→ Biometric Authentication.

→ Common biometric techn.

* Facial recognition.

* Finger print recognition.

* Hand geometry.

* Voice recognition.

* Signature verification.

Types of Authentication

→ Single - Factor Authentication (SFA)

* This relies on one form of verification typically a username & password.

→ Two - Factor Authentication

* Two - Factor Authentication requires two different authentication factors.

→ Password, OTP.

→ Multi Factor Authentication

* Two or more independent factors for identity verification.

Applications

- User Login systems.
- Banking & Financial Systems.
- Email & cloud services.

Client Based Access

* This is a cloud security mechanism that restricts access to cloud resources based on the client device or application requesting the service.

CBAC working

* The cloud system detects the device attempting access using methods such as device IDs (Client Identification).

- Device Authentication
- Security Posture Check
- Policy Evaluation
- Authorization
- Continuous Monitoring

Advantages

- Stronger security
- Reduces Insider threats
- Supports remote work
- Compliance support

API based Access

* API access refers to the ability to obtain access to & interact with an application programming interface to retrieve data or perform actions within a system.

* API acts as a secure messenger that allows two applications to communicate.

Types of API Access

- Public ⇒ Available to any developer
- Partner ⇒ Shared only with authorized business partners who have completed an onboarding ~~experience~~ process.
- Internal ⇒ Hidden from external users, & used only within an organization.
- Composite ⇒ Combines multiple API's into a single endpoint

API Based Working process

1.) Request ⇒ A client application sends a message to a specific URL.

* This includes GET, POST (create),
(retrieve)

PUT (update), or DELETE (remove).

2.) Authentication

3.) Processing

4.) Response ⇒ Server sends back the requested information.

Ex: - Weather data API

Geocoding API

Payment on a e-commerce website

Cloud Pricing Models

* This determines how you pay for the cloud resources & services.

* Generally Cloud infrastructure is pay-as-you-go pricing model

ON - Demand Pricing

* We pay only for what we use, with no long-term commitment.

Ex:- AWS on-demand Pricing, you are billed per second or per hour, depending on the service.

→ GCP On-Demand Pricing also works similarly. You are billed per second for virtual machines, storage, and other services.

→ Azure On-Demand Pricing You are billed per second or minute for services like virtual machines, storages & databases.

Reserved instances

* Reserved instances are a great way to save money if you know how much capacity you need.

* AWS offers Reserved instances for

Services like EC2.

- GCP calls its reserved options as Committed Use Discounts.
- Azure offers Reservations for its virtual machine instances.

Spot Instances

* These are unused cloud resources offered at highly discounted prices.

Savings Plans

- AWS = Offers savings plans with upto 72% discount.
- GCP = Provides committed use Discounts upto 57%.
- Azure = offers upto 65%.

Cloud Cost Optimization

